

Technical Document #642: Natural Language Processing - Comprehensive Overview

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Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Word embeddings represent words as dense vectors in continuous space.
- Language models predict the probability of a sequence of words.
- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.